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Dividend Policy at Linear Technology

It was April 2003 and Paul Coughlan was pulling together his notes for Linear Technology's board meeting the following day. As chief financial officer of the Silicon Valley semiconductor company, Coughlan was responsible for making a recommendation about whether or not Linear should increase its dividend this quarter. Coughlan and Linear's CEO Robert Swanson were pleased with the company's third-quarter financials for fiscal year 2003, but sales and net income still remained substantially below Linear's record levels set in 2001. In addition, the technology industry was still emerging from a recessionary environment and it was unclear how strong business would be for the remainder of the year.

Linear Technology Corporation

Headquartered in Milpitas, California, Linear was founded in 1981 by Robert Swanson. Under his leadership, the company focused on designing, manufacturing, and marketing integrated circuits (semiconductors) that were used in various electronic applications such as cellular telephones, digital cameras, complex medical devices, and navigation systems. Linear's customers spanned numerous industries and no single customer accounted for more than 5% of its business. In 2002, the communications industry accounted for 33% of Linear's business, computers 27%, automotive 6%, and the remaining 34% was spread across many different applications.

Linear focused on the analog segment within the broader semiconductor industry. Analog products were custom-designed for each application and were used to perform functions such as regulating power needs in complex electronics such as cell phones and digital cameras. Competition among analog companies was based in part on hiring and retaining top engineers who would continuously develop high-performance products. Capital investment in new fabrication facilities ("fabs") was also important, although the cost of a new analog fab (~\$200 million) was substantially below that of a digital fab, such as those being built by Intel (~\$2 billion). In addition, analog fabs could be used for 10-plus years, while digital fabs often become obsolete within three to five years.¹ Research and development expenses were also modest, peaking at \$102 million in fiscal year (FY) 2001. Within the analog segment of the industry, Linear competed with Maxim, Analog Devices, and

¹ Goldman Sachs, "Technology: Semiconductors," February 21, 2003, p. 91.

Professor Malcolm P. Baker and Alison Berkley Wagonfeld, Executive Director of the HBS California Research Center, prepared this case. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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National Semiconductor. Linear was the seventh-largest company, by market capitalization, in the Philadelphia Stock Exchange Semiconductor Index (SOX).

Financial History

Linear went public on the NASDAQ in 1986. The company split its stock four times since its IPO. (See **Exhibit 1** for Linear's stock price relative to the S&P 500. Volatility between 1998 and 2002 was 56%.) Coghlan was hired as the CFO shortly after the company went public.

Shareholders had rewarded Linear for solid growth and steady margins over the years. Linear's sales, gross profit and net income peaked in FY 2001 when technology spending worldwide was at an all-time high.² Business slowed down considerably in FY 2002, and Linear finished the year with net income of \$198 million relative to \$427 million from the previous year. (See **Exhibit 2** for selected income statement and cash flow data for Linear from 1992–2003.) In spite of the major drop in sales, Linear held its margins through cost cutting and work shutdowns. Unlike most companies in the semiconductor industry, Linear had established a variable cost structure early in its existence. A large fraction of employee compensation was profit sharing; in 2001, some employees received a 70% increase in their total compensation, while in 2002, profit sharing was set at a much lower level. Employee stock options were also a major component of compensation for Linear employees. These forms of variable compensation helped Linear maintain a positive cash flow in down years.

Financials for the first three quarters of fiscal year 2003 indicated some growth. Sales for the first half of FY 2003 were \$287 million as compared to \$241 million for the first half of FY 2002. In the company's announcement of its FY 2003 2Q numbers, Swanson had estimated that sales and profits would grow 3% to 7% over the next quarter. As Coghlan looked at the numbers for FY 2003 3Q, which would be announced the following week, he was pleased to see that the growth was in line with estimates from the previous quarter. Nevertheless, the numbers were still far below the 3Q financials for fiscal year 2001 and management did not see a clear path to reaching those levels again in the next year. In addition, the United States had just declared war on Iraq and it was unclear how the war would affect the economy in the U.S. and abroad. Linear was looking for new business opportunities in Asia, but the company was careful not to sacrifice its margins for top-line sales growth.

Dividend Policy

Linear announced its first dividend on October 13, 1992. Coghlan explained:

We knew that the analog market was a good business and that Linear was well positioned in the industry. We had been cash flow positive since our IPO and we wanted to show investors that buying shares in Linear was not as risky as buying shares in most technology companies. In addition, we believed that offering a dividend would give us access to a new set of investors that had income goals in addition to growth goals.

The quarterly dividend was initially set at \$0.05 per share. This amounted to \$8.3 million, or 15% of FY 1994 earnings. According to Coghlan, "We initially set the dividend at a relatively low level because we were aware that investors respect companies for paying a dividend, but hammer those firms that reduce or stop paying dividends. We were thoughtful about what payout ratio we could sustain over time."

² Linear's fiscal year ends in June.

Linear had paid dividends each quarter since then. (See **Exhibit 3** for quarterly dividend data for Linear since 1992.) Linear's management and board debated about increasing the dividend in 2002 amidst the company's drop in sales and earnings, and ultimately they decided to continue with the increase in order "to validate [Linear's] ability to be profitable and cash flow positive even in tough times."³ Coghlan expanded:

We knew that our payout ratio would be closer to 25% to 30%, but we continued to remain confident about our business prospects and our cost structure. We believe that investors put us in a slightly higher category because of our consistent dividends and we did not want to lose that edge. In addition, with a 1% dividend yield in such a low-interest-rate environment, some investors think of us as a source of income with a bonus of growth potential.

Linear's management also believed that given the company's positive cash flow, additional money could be returned to shareholders in the form of share repurchases. In some years Linear allocated substantially more cash to share buybacks than to dividends. (See **Exhibit 3** for quarterly repurchase data for Linear since 1992.) Coghlan explained, "One of the primary reasons that we buy back stock is to offset the exercise of employee stock options." Another factor was market conditions. Coghlan considered the valuation of Linear's shares and the return on cash balances: "For the last few years interest rates have been so low that our cash was not earning much interest income in the high-grade securities in which we kept it. This encouraged us to use our cash balances to buy back even more shares."

Despite the dividends and repurchases, Linear had accumulated a large cash balance over the years. There were no plans to make acquisitions. Although Linear was a star performer in its own segment, Coghlan had no illusions about running someone else's business. Yet, in March of 2003, cash and short-term investments stood at over \$1.5 billion. The cash was managed very conservatively, in short-term debt securities. Interest income in 2002 amounted to \$52 million. Some of Linear's investors had urged Coghlan to return some of this cash, perhaps in the form of a special dividend, but he had the impression that this sentiment was not widespread.

Janus Capital was the largest single holder of Linear stock. (See **Exhibit 4** for institutional ownership data for Linear in 2003. No individual held more than 1% of shares outstanding.) Blaine Rollins was the lead portfolio manager of the Janus Fund, and Linear was his third-largest position, representing over 5% of the fund's portfolio. He was comfortable with Coghlan's approach:

Linear has always had strong cash flow. I liked that they bought back stock, particularly when the price was weak. Although they might have a high dividend for a tech company, it still seems "token" to me given how much cash they have. I understand why they pay dividends, however, as there still are many mutual funds and European investment firms that will only buy stocks with dividends.

Trends in Dividend Policy

Dividend payers have tended to be larger, more consistently profitable firms, with slower growth in sales and assets and fewer future growth opportunities. For example, slow growing and profitable utilities pay a much larger fraction of their earnings in dividends than faster growing pharmaceutical, biotechnology, and information technology firms. (See **Exhibit 5** for the dividend policy of the S&P 500 firms over the last decade.) In addition, there has been a general trend over time away from

³ Linear Technology 2002 Annual Report, p. 2.

dividends. In 1978, over two-thirds of the NYSE, AMEX, and NASDAQ firms paid a regular dividend. By 1999, only 21% were dividend payers. Part of this could be attributed to changing firm characteristics. In the 1980s and 1990s, firms were going public earlier in their lifecycle, with considerably more growth opportunities than current profits. This type of firm had never paid dividends. But this was at most half of the story. Even large and profitable firms were initiating dividends at a much slower rate.⁴ In large part, technology firms were driving this trend.

Profitability and investment opportunities Some market observers felt that many technology companies were only “just beginning to reach the stage at which paying dividends becomes possible. As a company establishes a regular and predictable cash flow from operations, then it usually considers whether to return some of that cash to its shareholders.”⁵ Repurchases were uncommon before the mid-1980s.⁶ Unlike dividends, repurchases are more flexible, in part because they are “discretionary—and once announced, may not always be implemented.”⁷ But some analysts said that having less flexibility and “less cash on hand to throw into new projects or acquisitions may not be all bad, forcing a measure of restraint on companies keen on acquisitions.”⁸ A related issue is how a company wants investors to perceive its growth opportunities. As David Readerman, growth stock strategist at Thomas Weisel Partners explained, “Executives will be cautious about signaling to the market that their growth rate is slowing, which is what a dividend is usually seen as doing.”⁹

Executive stock options Even large, stable, and profitable technology companies tended not to pay dividends. One factor seemed to be the heavy use of stock option compensation in this sector. With the bulk of their pay in options, managers may have little incentive to pay dividends, because “options become more valuable as the stock price rises, but do not benefit from dividend payments.”¹⁰ (See **Exhibit 6** for the breakdown of executive compensation into salary, bonus, and stock options for select technology companies.) Also, as a practical matter, “many tech companies use their available cash to buy back shares to offset the dilution caused when employees exercise their options. If companies instead pay out their cash in the form of dividends, then their shares outstanding will continue to rise, which means that their earnings per share will suffer—even as a dividend soaks up ever more cash.”¹¹

Taxes In the United States, corporate profits have been taxed twice, at the corporate level and the investor level. In January 2003, President George W. Bush proposed a radical change to the taxation of investment income. The Bush plan was to eliminate taxes on dividends, if they were paid out of earnings that had already been taxed. The initial proposal also included an adjustment to the capital gains tax. Each investor’s tax basis would rise by retained (and taxed) earnings per share. Later in the spring, the House of Representatives had put forward a less costly bill, where both

⁴ See Eugene Fama and Kenneth French, “Disappearing Dividends: Changing Firm Characteristics or Reluctance to Pay,” *Journal of Financial Economics* 60, pp. 3–44, 2001.

⁵ Tam Harbert, “Dividend Dilemma,” *Electronic Business*, March 1, 2003. (Printed from <www.reed-electronics.com> on August 7, 2003.)

⁶ In 1982, the Securities Exchange Commission (SEC) adopted Rule 10b-18. Under this rule, a firm meeting certain “safe harbor” conditions would avoid claims that a repurchase was being used to manipulate its share price.

⁷ Suzanne McGee, “Are Dividends in Your Tech Fund’s Future?” *Standard & Poor’s*, March 14, 2003.

⁸ David R. Baker, “Tech Firms Taking Stock of Dividend Tax Cut Plan,” *San Francisco Chronicle*, January 8, 2003.

⁹ Suzanne McGee, “Are Dividends in Your Tech Fund’s Future?” *Standard & Poor’s*, March 14, 2003.

¹⁰ Floyd Norris, “Growing Number of Companies Choose Not to Offer Dividends,” *The New York Times*, January 4, 2000.

¹¹ Ken Brown and Jesse Eisinger, “Tech Dividends Deserve Closer Look,” *The Asian Wall Street Journal*, January 13, 2003.

dividends and capital gains would be taxed at a rate of 15%, still a dramatic reduction from the highest ordinary income tax rate of 38.6%.¹² (See **Exhibit 7** for historical tax rates.) Of course, not every investor paid taxes at the highest rates. For example, most endowments are tax-exempt. But some academic studies have found a positive link between dividend yields and future returns, suggesting that investors demand compensation for the tax burden.¹³ Likewise, the pricing of municipal and corporate bonds suggested that investors as a group preferred tax-exempt distributions to taxable ones. The majority of municipal bonds are exempt from federal and state income taxes. (See **Exhibit 8** for the pricing of municipal and AAA-rated bonds.)

Before the proposals, Rollins “tended to prefer companies with minimal or no dividends, as I always felt that we were giving more to the government than necessary. I encouraged my companies to buy back stock instead.” But if the proposal went through, he would “encourage companies to use both mechanisms. For companies with strong excess cash flow such as Linear, I would suggest saving a third of the cash for a ‘rainy day’ and sharing the other two-thirds with investors, split equally between dividends and buybacks.”

Market valuations The value of dividend payers and nonpayers varied over time. (See **Exhibit 9** for a comparison of the market-to-book ratios.) Since the middle of 2000, the returns of dividend payers had outstripped those of nonpayers by a wide margin. The semiconductor industry was no exception. (See **Exhibit 10** for a comparison of the returns in the SOX Index.) These returns, and the wake of the recent corporate scandals at Enron and Worldcom, had ignited interest in dividend-based investing. In late 2002, a *Fortune* article explained that “going forward, the theory holds, a growing proportion of returns will come from income rather than from capital gains, and investors will flock to dividend-producing shares.”¹⁴

This was a complete reversal from early 2000, when nonpayers were the star performers. Just two years earlier, the *New York Times* reported in a front-page article:

A growing portion of corporate America appears to be concluding that dividends are no longer needed to attract investors. . . . The only similar trend occurred in the late 1960’s, another time that small technology companies were all the rage and the market for new issues was red hot. . . . The most likely explanation would seem to be the most obvious. Investors, after seeing year after year of huge capital gains, no longer see much of a need for dividends as an assured return if the market declines.¹⁵

For Rollins, this shift in the appetite for dividends worked in harmony with the tax changes. He commented, “Without a tax change, I would have said ‘No’ [to Linear raising its dividend]. With a change, I am comfortable with their consistent increase. I see it as having a bird in the hand versus two in the bush. Linear’s dividend lowers its equity risk premium in today’s market, which is something we like.”

¹² The highest personal tax rate was 38.6%; the capital gains tax rate was 20%; and the highest corporate tax rate was 35%.

¹³ For example, see Robert Litztenberger and Krishna Ramaswamy, “The Effects of Personal Taxes and Dividends on Capital Asset Prices: Theory and Evidence,” *Journal of Finance* 37, 429–443.

¹⁴ Richard Teitelbaum, “Investor’s Guide 2003: Playing the Dividend Market,” *Fortune Magazine*, December 3, 2002.

¹⁵ Floyd Norris, “Growing Number of Companies Choose Not to Offer Dividends,” *The New York Times*, January 4, 2000.

Dividend Policy at Technology Firms

Of the 16 companies on the SOX index, six paid dividends. Coghlan focused on Intel and Maxim as benchmarks. The other dividend payers in the SOX index were two older companies, Motorola and Texas Instruments, and France-based ST Microelectronics. (See **Exhibit 11** for selected information about the companies included in the SOX.) Also, recent events at Microsoft and Cisco had caught Coghlan's attention.

Intel

Intel had paid a dividend for over 10 years. It first initiated a quarterly dividend of \$0.10 per share in September 1992, just one month before Linear's initiation. According to the *New York Times*:

Analysts welcomed the dividend, which they said should make Intel shares less volatile and broaden the company's investor base, because some mutual funds will only invest in stocks that pay dividends . . . the dividend is a statement of the confidence of Intel in the face of numerous challenges to the monopoly it once enjoyed in the market for microprocessors . . . it shows the company's desire to be considered a more mature, though still growing, concern.

Richard Shaffer, publisher of the *Technologic Computer Letter*, took a skeptical view. "When a high tech company starts paying a dividend, that indicates the company believes that shareholders can make higher returns going elsewhere. If I were an Intel shareholder, I'd rather they put my 40 cents a year back into innovation."¹⁶

Intel split its stock several times since 1992 and as of spring 2003, Intel paid \$0.02 per share on a quarterly basis. Intel had not increased its dividend since 2000. (See **Exhibit 12** for Intel financials).

Maxim

Intel had the largest cash position of all the semiconductor companies, but several others (in addition to Linear) paid dividends as well. Two companies—Maxim Integrated Products (Maxim) and Microchip Technology (Microchip)—announced in fall 2002 that each would start paying a quarterly dividend of \$0.02 per share. Maxim and Linear were similar in size and financial performance, and they both competed in the analog segment. (See **Exhibit 12** for Maxim financials). Rollins, who also held a large chunk of Maxim stock, felt "Maxim's recent decision to start offering a dividend was driven in large part by the desire for more European firms to buy its shares."

Microsoft

Just before President Bush announced his tax proposal, software leader Microsoft declared its first quarterly dividend of \$0.04 per share, payable on March 7, 2003.¹⁷ For the quarter ending in December 2002, Microsoft's earnings were \$0.47 per share, slightly higher than forecast. (See **Exhibit 12** for Microsoft financials.) Historically, Microsoft favored returning money to shareholders in the form of stock buybacks, but according to Microsoft CFO John Connors, "Because we have been able to resolve some legal claims, including a proposed \$1.1 billion class-action lawsuit settlement with

¹⁶ Lawrence M. Fisher, "Intel to Pay a Dividend, It's First Ever," *The New York Times*, September 18, 1992, p. 3.

¹⁷ Microsoft also announced a plan to split its shares, so the quarterly dividend would be \$0.02/share post split.

California consumers, we are more confident about paying a 'starter dividend.'"¹⁸ He said that "the company would monitor ongoing R&D and legal costs, and could always increase the dividend later as circumstances permitted."¹⁹ Some analysts praised the decision, citing that Microsoft was finally returning some of its \$38.7 billion of cash to shareholders. Others said the amount was "miniscule," and that Microsoft was "just throwing a token out there to get people enthused about buying its stock."²⁰

Cisco

After Microsoft's announcement, many wondered if Cisco, the world's leader in networking and communication products, would follow suit, announcing a dividend to distribute a portion of its \$12.7 billion in cash. (See **Exhibit 12** for Cisco financials.) At Cisco's annual meeting in November 2002, shareholders were given the opportunity to vote on a proposal to initiate a dividend. The proposal was defeated. As reported by *InformationWeek*:

The board of directors recommended voting against the shareholder initiative, and since the board was re-elected by a 95% mandate of the voting shareholders, it isn't too surprising that its recommendations would be followed. John Chambers, the popular CEO, has often stated that he would prefer seeing the company buy back stock or make acquisitions. This alone would make many shareholders vote against the proposition. And let's not forget that part of this argument against paying dividends is that they are taxed twice—once at the corporate level and then a second time at the shareholder level.²¹

Since the vote and proposed tax law changes, the company had continued to assert that it would prefer to use its cash for investment opportunities or to buy back stock.

Dividend Decision for Linear

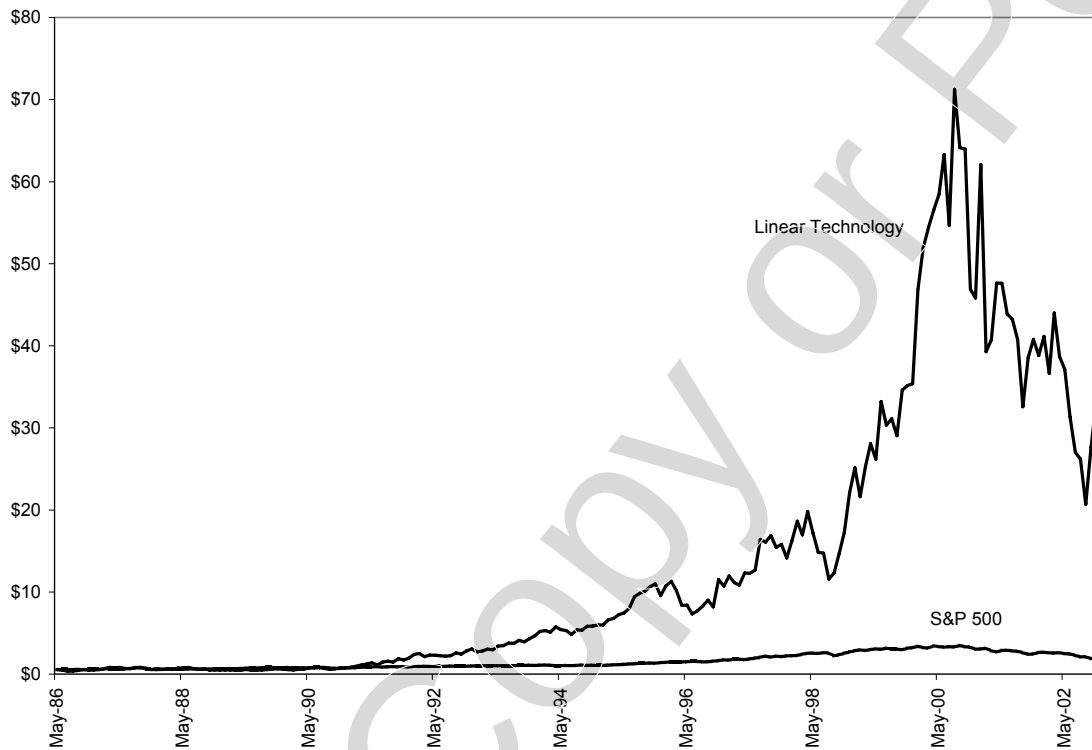
Coghlan thought about what he wanted to recommend to the board in light of Linear's own financial prospects as well as the broader technology and capital markets environment. He was aware that dividend decisions often gave signals to shareholders as to how management perceived the business. He also knew that increasing the dividend another penny would move the payout ratio to a level that was above most other technology companies.

¹⁸ Helen Jung, "Microsoft Dividend Ushers in New Era," *Associated Press*, January 20, 2003.

¹⁹ Duncan Stewart, "Microsoft Dividend Not Good Enough: Yield is Pathetic, Even Given Low Interest Rates," *Financial Post*, January 18, 2003.

²⁰ Quote by Bob Cordiak, RBC Dain Rauscher, Inc., in an article by Bill Deener and Alan Goldstein, "Microsoft Plans to Pay its First Dividend," *The Dallas Morning News*, January 17, 2003.

²¹ William Schaff, "Taking Stock: Cisco Investors Miss Out on Dividend Vote," *InformationWeek*, November 25, 2002.

Exhibit 1 Linear Stock Price History, 1986–2002

Source: Adapted from the Center for Research on Security Prices, adjusted for splits and dividends.

Exhibit 2 Selected Linear Financial Data in Millions of Dollars (Except Share Data), 1992–2003

June Fiscal Year	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003 ^a
Income Statement												
Sales	119.4	150.9	200.5	265.0	377.8	379.3	484.8	506.7	705.9	972.6	512.3	440.8
-Cost of Goods Sold	49.5	57.0	67.6	83.3	106.8	109.8	137.8	139.8	179.0	231.1	144.7	114.6
-Research and Development Expense	12.3	14.8	18.4	23.9	31.1	35.4	46.2	54.7	78.3	102.5	79.8	67.0
-Other Expenses	19.1	23.3	28.0	29.4	36.0	29.6	29.6	26.5	31.4	28.4	9.4	18.9
Income Before Taxes	38.5	55.7	86.5	128.5	203.9	204.5	271.3	285.7	417.3	610.7	278.4	240.3
-GAAP Income Taxes	13.5	19.3	29.7	43.8	69.9	70.2	90.4	91.4	129.4	183.2	80.7	69.7
Net Income	25.0	36.4	56.8	84.7	134.0	134.4	180.9	194.3	287.9	427.5	197.6	170.6
Common Shares Outstanding (Split-Adjusted)	280.2	285.4	290.5	294.3	298.6	303.8	307.3	307.5	315.2	318.9	316.2	312.4
Earnings Per Share (Split-Adjusted)	\$0.09	\$0.13	\$0.20	\$0.29	\$0.45	\$0.44	\$0.59	\$0.63	\$0.91	\$1.34	\$0.63	\$0.55
Dividends Paid—Preferred	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Dividends Paid—Common	0.0	5.3	8.3	9.8	11.9	15.0	18.3	22.1	28.0	41.2	54.0	47.0
Share Price, Fiscal Year Close (Split-Adjusted)	\$2.36	\$3.61	\$5.50	\$8.25	\$7.50	\$12.94	\$15.08	\$33.63	\$63.94	\$44.22	\$31.43	\$30.87
Cash Flow Statement												
Net Income	25.0	36.4	56.8	84.7	134.0	134.4	180.9	194.3	287.9	427.5	197.6	170.6
+Depreciation and Amortization	4.5	5.8	6.3	8.6	10.3	12.4	20.1	22.0	25.0	35.8	46.3	33.5
-Capital Expenditure	9.8	7.6	16.2	22.1	70.4	21.9	24.4	39.1	80.3	127.9	17.9	9.8
-Working Capital Investments	7.2	2.6	6.8	-4.5	-9.1	21.2	-7.4	-9.6	-26.6	19.2	42.0	14.9
+Tax Adjustment ^b	6.4	2.8	6.1	6.2	11.3	25.3	58.6	54.6	102.9	115.5	55.2	0.7
Operating Cash Flow	19.0	34.8	46.3	81.8	94.3	129.1	242.6	241.3	362.0	431.7	239.3	180.1
+Stock Issuance	11.7	9.1	13.7	7.6	12.7	18.5	26.6	38.3	54.8	52.7	39.3	27.4
-Stock Purchases	0.7	1.2	1.3	6.1	22.9	11.6	56.5	108.7	0.0	69.8	221.6	165.7
-Dividends Paid	0.0	5.3	8.3	9.8	11.9	15.0	18.3	22.1	28.0	41.2	54.0	47.0
-Other Items ^c	0.0	4.7	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-18.4
Net Cash Flow	30.0	32.6	48.9	73.4	72.3	121.0	194.5	148.8	388.9	373.4	3.0	13.2
Cash and Short-Term Investments	95.3	127.9	176.8	250.2	322.5	443.4	637.9	786.7	1,175.6	1,549.0	1,552.0	1,565.2

Source: Adapted from Compustat.

^aFirst three quarters of FY2003.^bThe difference between the exercise price and the market value of LLTC stock could be expensed for tax purposes, leading to large tax adjustments on the cash flow statement.^cOther Items includes long-term investments and acquisitions (other than capital expenditure), extraordinary items, and other adjustments to net income.

Exhibit 3 Linear Payout Policy, 1992–2003

Fiscal Quarter	Stock Re-Purchase (\$M)	Dividends Per Share ^a (\$)	Stock Splits (Date)	Share Price ^a (\$)	Shares ^a (M)	Market Reaction			
						Dividend ^b	Earnings ^b	LLTC Return ^c (%)	SOX Return ^c (%)
Q1 93		0.00000		2.70	281.3	13-Oct	14-Oct	-1.6	
Q2 93		0.00625	2:1 (11/3)	3.19	283.1		12-Jan	-1.0	
Q3 93	1.2	0.00625		3.16	283.4		13-Apr	11.0	
Q4 93		0.00625		3.61	285.4		20-Jul	-1.2	
Q1 94		0.00625		4.28	294.1	12-Oct	12-Oct	11.7	
Q2 94		0.00750		4.84	287.7		19-Jan	0.6	
Q3 94	0.5	0.00750		5.28	289.6		19-Apr	-4.1	
Q4 94	0.8	0.00750		5.50	290.5				
Q1 95	6.1	0.00750		5.54	290.0	17-Oct	17-Oct	2.2	
Q2 95		0.00875		6.19	290.5		18-Jan	4.7	0.5
Q3 95		0.00875		7.00	293.2		18-Apr	1.8	-5.0
Q4 95		0.00875		8.25	294.3	25-Jul	27-Jul	11.7	0.2
Q1 96		0.01000	2:1(8/11)	10.38	295.2		17-Oct	14.3	7.3
Q2 96	3.5	0.01000		9.81	296.3		17-Jan	10.6	3.6
Q3 96	5.0	0.01000		10.44	299.3		16-Apr	-14.3	4.6
Q4 96	14.4	0.01000		7.50	298.6	23-Jul	23-Jul	-1.4	-5.8
Q1 97	11.6	0.01250		9.22	297.5		16-Oct	-11.0	-1.3
Q2 97		0.01250		10.97	299.5		15-Jan	11.4	2.3
Q3 97		0.01250		11.06	301.6		16-Apr	-2.3	-0.7
Q4 97		0.01250		12.94	303.8	23-Jul	22-Jul	8.3	3.9
Q1 98		0.01500		17.19	306.0		14-Oct	5.4	-4.0
Q2 98	50.5	0.01500		14.41	303.0		13-Jan	6.6	6.4
Q3 98	5.9	0.01500		17.25	306.0		14-Apr	13.3	6.8
Q4 98		0.01500		15.08	307.3	21-Jul	21-Jul	-15.5	-5.0
Q1 99	100.0	0.01750		12.50	300.7		13-Oct	11.9	9.0
Q2 99	8.8	0.01750		22.39	301.3		12-Jan	7.9	3.1
Q3 99		0.01750	2:1(1/29)	25.63	306.1	13-Apr	13-Apr	0.3	-6.2
Q4 99		0.02000		33.63	307.5		20-Jul	-10.7	-6.4
Q1 00		0.02000		29.39	308.5		12-Oct	-3.5	-1.4
Q2 00		0.02000		35.78	311.1		18-Jan	6.1	-0.5
Q3 00		0.02000	2:1(3/6)	55.00	313.1	18-Apr	18-Apr	15.2	13.5
Q4 00		0.03000		63.94	315.2		25-Jul	-2.0	-4.5
Q1 01		0.03000		64.75	316.6		17-Oct	-18.3	-14.6
Q2 01	52.3	0.03000		46.25	316.0		16-Jan	8.8	1.0
Q3 01	11.0	0.03000		41.06	317.8	17-Apr	17-Apr	8.5	7.2
Q4 01	6.5	0.04000		44.22	318.9		24-Jul	-11.3	-3.6
Q1 02	114.8	0.04000		32.80	316.3		16-Oct	-10.1	-9.5
Q2 02	3.0	0.04000		39.04	317.2		15-Jan	-1.1	-6.5
Q3 02	37.3	0.04000		44.22	317.1	16-Apr	17-Apr	4.9	3.7
Q4 02	66.5	0.05000		31.43	316.2		23-Jul	0.5	-2.9
Q1 03	125.0	0.05000		20.72	311.8		15-Oct	4.3	-0.3
Q2 03	1.5	0.05000		25.72	313.4		14-Jan	-2.4	-4.2
Q3 03	39.2	0.05000		30.87	312.4				

Source: Adapted from Compustat; the Center for Research on Security Prices; and Thomson Financial Datastream.

^aAdjusted for splits.

^bAnnouncement date. The quarterly earnings announcement date for Q4 1994 is not available from Compustat.

^cReturn on the day before, the day of, and the day after the earnings announcement. The SOX index begins in October 1994.

Exhibit 4 Top 10 Institutional Shareholders, 2003

Institution	Shares	Percent of Shares Outstanding
Janus Capital Management	34,151,384	10.91
Capital Research and Management Company	23,807,150	7.61
State Farm Mutual Automobile Insurance	14,599,700	4.66
Fidelity Management & Research (FMR)	10,902,153	3.48
Barclays Bank	10,561,160	3.37
Oak Associates	9,754,731	3.12
State Street	9,468,226	3.03
Northern Trust	7,473,564	2.39
AXA Financial	7,429,952	2.37
Massachusetts Financial Services	7,309,046	2.34

Source: Adapted from Thomson Financial Spectrum Database.

Exhibit 5 Dividend Policy for the S&P 500, Annual Data from 1993–2002

Sector	Dividend Yield	Earnings Yield	Return on Book Equity
Utilities	4.8%	7.1%	11.4%
Telecommunications Services	2.3%	5.2%	14.4%
Materials	2.2%	5.4%	12.2%
Energy	1.8%	5.2%	9.8%
Financials	2.3%	7.0%	16.5%
Industrials	1.8%	5.6%	15.2%
Consumer Staples	1.8%	4.6%	21.1%
Consumer Discretionary	1.4%	5.4%	15.5%
Health Care	0.9%	3.9%	16.9%
Information Technology	0.3%	3.6%	14.9%
Weighted Average	1.8%	5.4%	15.1%

Source: Adapted from *Goldman Sachs Portfolio Strategy* (July 1, 2003 Research Report).

Note: The sample includes S&P 500 firms in 2002. Not all firms have data back to 1993.

Exhibit 6 Cumulative Executive Compensation, 1993–2002

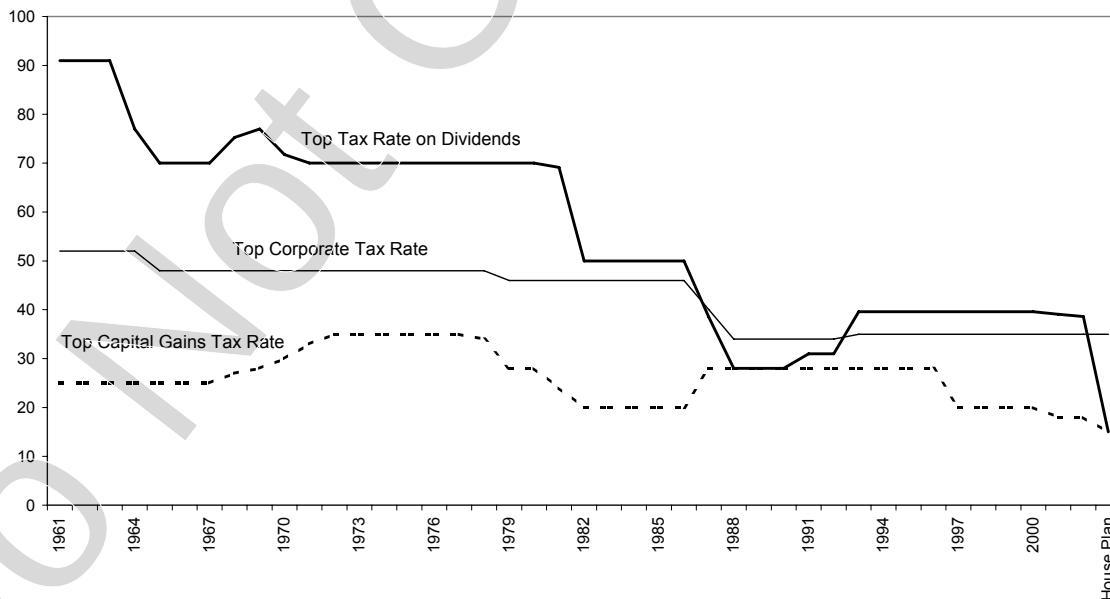
Company	Name (CEO Year)	Salary (\$M)	Bonus (\$M)	Stock Options Grants ^a (000)	Stock Options Exercise ^{a,b} (000)	Value Realized from Exercise (\$M)	Options Owned ^c (000)	Shares Owned ^c (000)	Common Shares (M)
Linear Technology	Robert Swanson	2.6	12.6	3,000	1,297	29.1	1,003	530	316
Cisco Systems	John Chambers (1995)	2.3	3.3	45,840	43,090	344.9	35,400	2,047	7,303
Intel	Craig Barrett (1998)	4.3	17.8	5,005	10,752	196.6	5,005	2,732	6,575
Maxim	John Gifford	2.7	12.0	10,913	10,522	206.3	4,946	2,047	320
Microsoft	Steven A. Ballmer (2000)	3.5	2.0	0	0	0.0	0	410,968	5,359

Source: Adapted from Execucomp.

^aAdjusted for splits.

^bIncludes the exercise of options granted prior to 1993.

^cAs reported in the proxy statement in 2002. In 2002, Robert Swanson and John Chambers forfeited 700,000 options and 2 million options, respectively.

Exhibit 7 Tax Rates, 1960–2002

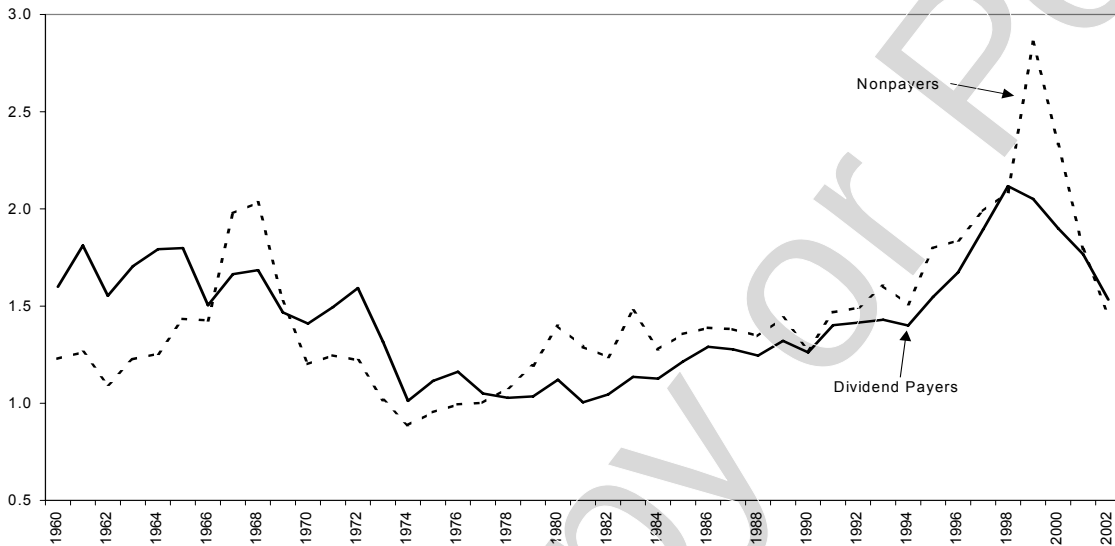
Source: Congressional Joint Committee on Taxation, as reported at <www.house.gov/jct/x-6-01.pdf> on June 30, 2003.

Exhibit 8 Municipal and Corporate Bond Yields

Year	Bloomberg 1-year AAA Corporate Debt	Bloomberg 1-year AAA Muni Debt
Annual Average Month-End Yield (%):		
1995	6.25	4.17
1996	5.89	3.81
1997	5.94	3.92
1998	5.44	3.68
1999	5.73	3.64
2000	6.71	4.49
2001	3.87	2.94
2002	2.37	1.87

Source: Adapted from Bloomberg.

Exhibit 9

Market-to-Book Ratio^a of Dividend Payers and Non-Payers, 1960–2002

Source: Adapted from Malcolm Baker and Jeffrey Wurgler, "A Catering Theory of Dividends," *Journal of Finance*, forthcoming; Compustat.

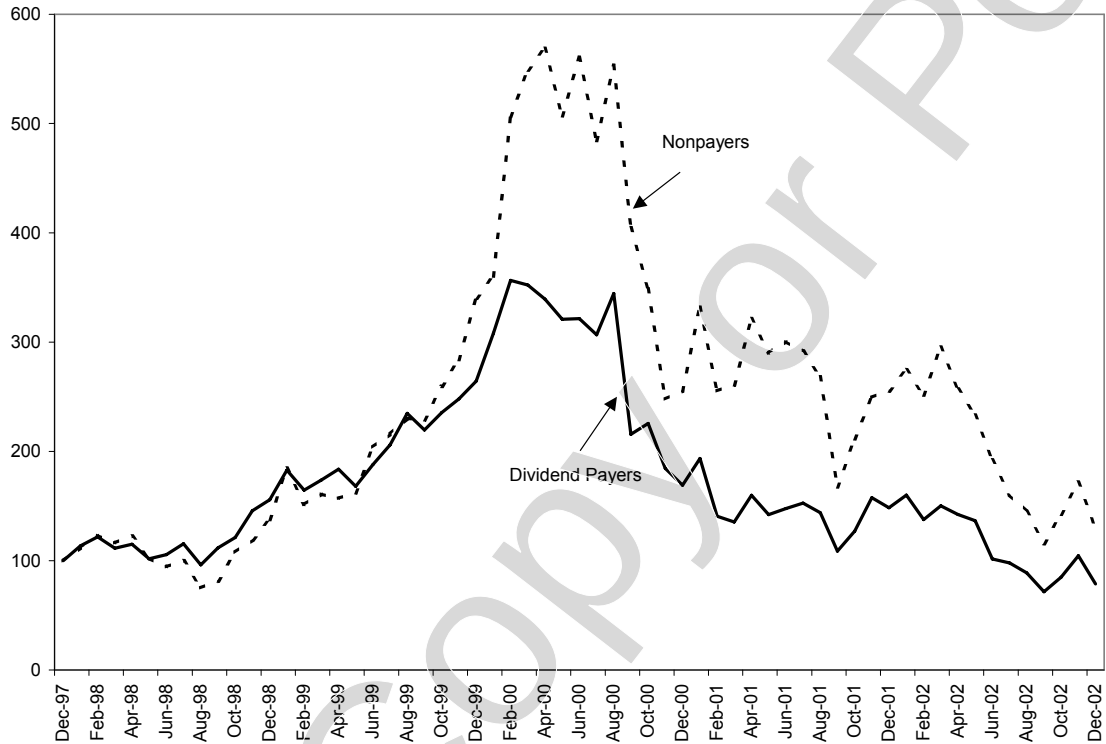
^aRatio of market equity plus book debt dividend by book assets.

Dividend Initiation Rates^b (solid line) and Initiations (dashed line), 1962–2000

Source: Adapted from Malcolm Baker and Jeffrey Wurgler, "A Catering Theory of Dividends," *Journal of Finance*, forthcoming; Compustat.

^bInitiations divided by surviving nonpayers. The Center for Research on Security Prices coverage of NASDAQ begins in 1973, increasing the sample of nonpayers.

Exhibit 10 Price Index of Dividend Payers (solid line) and Nonpayers (dashed line) on the Philadelphia Stock Exchange Semiconductor Index (SOX), 1998–2002



Source: Adapted from the Center for Research on Security Prices, market value weighted.

Exhibit 11 Data on Companies in the Semiconductor Index (SOX) in Millions of Dollars (Except Share Data), 2002

Company	Ticker	Share Price ^a	Shares (Millions)	Net Income	Cash Flow ^b	Cash ^c	Long-Term Debt	Dividends	Stock Re-purchases	Dividend Initiation Date
Advanced Micro Devices	AMD	6.18	344.5	-1,303.0	-794.1	1,037.7	1,779.8	0.0	0.0	Apr-95 ^d
Altera	ALTR	13.54	383.5	91.3	237.8	942.7	0.0	0.0	139.5	
Applied Materials	AMAT	12.58	1,648.0	269.0	75.1	4,929.5	573.9	0.0	125.0	
Broadcom	BRCM	12.35	277.8	-2,236.6	-144.4	502.7	1.2	0.0	0.0	
Intel	INTC	16.28	6,575.0	3,117.0	4,426.0	12,587.0	929.0	533.0	4,014.0	Sep-92
Kla-Tencor	KLAC	35.94	189.8	216.2	215.4	673.3	0.0	0.0	123.2	
Linear Technology	LLTC	30.87	316.2	197.6	239.3	1,552.0	0.0	54.0	221.6	Oct-92
Lsi Logic	LSI	4.52	375.1	-292.4	115.1	990.0	1,241.2	0.0	0.0	
Maxim Integrated Products	MXIM	36.12	320.1	259.2	313.4	765.5	0.0	0.0	864.0	Oct-02
Micron Technology	MU	8.14	602.9	-907.0	-181.8	985.7	360.8	0.0	0.0	Sep-91 ^e
Motorola	MOT	8.26	2,315.3	-2,485.0	732.0	6,566.0	7,674.0	364.0	0.0	Nov-46
National Semiconductor	NSM	17.04	183.8	-33.3	49.8	915.4	19.9	0.0	0.0	
Novellus Systems	NVLS	27.27	149.1	22.9	188.1	1,019.7	0.0	0.0	78.2	
STMicroelectronics ^f	STM	18.90	887.5	429.0	718.0	2,564.0	2,797.0	36.0	115.0	May-99
Taiwan Semiconductor ^g	TSM	6.84	3,703.2	413.8	951.3	1,113.0	1,701.7	0.0	0.0	
Teradyne	TER	11.64	183.2	-718.5	-80.6	325.4	450.6	0.0	0.0	
Texas Instruments	TXN	16.37	1,730.6	-344.0	1,190.0	3,012.0	833.0	147.0	370.0	Apr-26
Xilinx	XLNX	23.41	339.0	125.7	299.0	675.6	0.0	0.0	60.8	

Source: Adapted from Compustat; Center for Research on Security Prices.

^aShare price on March 31, 2003.^bCompustat operating cash flow (Item 308) less capital expenditures (Item 128).^cCash and short-term investments.^dSpecial (non-recurring) dividend.^eMicron discontinued its dividend in 1996.^fSTMicroelectronics is an American Depository Receipt (ADR), based in France.^gTaiwan Semiconductor is an ADR, based in Taiwan.

Exhibit 12 Selected Financial Data in Millions of Dollars (Except Share Data), 1998–2002

	1998	1999	2000	2001	2002
CISCO (July Fiscal Year)					
Sales	8,458.8	12,154.0	18,928.0	22,293.0	18,915.0
Income Before Taxes	2,302.5	3,316.0	4,343.0	-874.0	2,710.0
-GAAP Income Taxes	952.4	1,220.0	1,675.0	140.0	817.0
Net Income	1,350.1	2,096.0	2,668.0	-1,014.0	1,893.0
+Depreciation and Amortization	327.3	486.0	863.0	2,236.0	1,957.0
-Capital Expenditure	414.8	584.0	1,086.0	2,271.0	2,641.0
-Working Capital Investments	-688.7	-937.0	-1,262.0	-5,078.0	-2,829.0
+Tax Adjustment	514.4	919.0	1,348.0	92.0	-92.0
Operating Cash Flow	2,465.7	3,854.0	5,055.0	4,121.0	3,946.0
+Stock Issuance	488.5	740.0	1,564.0	1,262.0	655.0
-Stock Purchases	0.0	0.0	0.0	0.0	1,854.0
-Dividends Paid	0.0	0.0	0.0	0.0	0.0
-Other Items ^a	2,538.3	4,269.5	3,110.0	4,001.0	-3,002.0
Net Cash Flow	415.9	324.5	3,509.0	1,382.0	5,749.0
Cash and Short-Term Investments	1,691.5	2,016.0	5,525.0	6,907.0	12,656.0
Common Shares Outstanding (Split-Adj)	6,250.3	6,542.0	7,138.0	7,324.0	7,303.0
Share Price, Fiscal Year Close (Split-Adj)	\$15.96	\$31.06	\$65.44	\$19.22	\$13.19
INTEL (December Fiscal Year)					
Sales	26,273.0	29,389.0	33,726.0	26,539.0	26,764.0
Income Before Taxes	9,137.0	11,228.0	15,141.0	2,183.0	4,204.0
-GAAP Income Taxes	3,069.0	3,914.0	4,606.0	892.0	1,087.0
Net Income	6,068.0	7,314.0	10,535.0	1,291.0	3,117.0
+Depreciation and Amortization	2,807.0	3,186.0	4,807.0	6,052.0	5,042.0
-Capital Expenditure	3,557.0	3,403.0	6,674.0	7,309.0	4,703.0
-Working Capital Investments	-31.0	180.0	2,912.0	-1,627.0	-358.0
+Tax Adjustment	285.0	1,015.0	397.0	-316.0	612.0
Operating Cash Flow	5,634.0	7,932.0	6,153.0	1,345.0	4,426.0
+Stock Issuance	2,127.0	543.0	797.0	762.0	681.0
-Stock Purchases	6,785.0	4,612.0	4,007.0	4,008.0	4,014.0
-Dividends Paid	168.0	366.0	470.0	538.0	533.0
-Other Items ^a	3,109.0	-665.0	438.0	-166.0	-477.0
Net Cash Flow	-2,301.0	4,162.0	2,035.0	-2,273.0	1,037.0
Cash and Short-Term Investments	7,626.0	11,788.0	13,823.0	11,550.0	12,587.0
Common Shares Outstanding (Split-Adj)	6,632.0	6,668.0	6,721.0	6,690.0	6,575.0
Share Price, Fiscal Year Close (Split-Adj)	\$29.64	\$41.16	\$30.06	\$31.45	\$15.57

Exhibit 12 (continued)

	1998	1999	2000	2001	2002
MAXIM (June Fiscal Year)					
Sales	560.2	607.0	864.9	1,576.6	1,025.1
Income Before Taxes	269.9	297.2	425.2	505.0	386.8
-GAAP Income Taxes	91.8	101.0	144.6	170.0	127.7
Net Income	178.1	196.1	280.6	334.9	259.2
+Depreciation and Amortization	15.1	17.6	19.7	90.9	56.3
-Capital Expenditure	109.4	54.3	176.1	336.5	90.4
-Working Capital Investments	-18.8	-34.8	-26.2	-235.6	48.4
+Tax Adjustment	83.5	57.7	157.3	148.3	136.8
Operating Cash Flow	186.1	251.9	307.7	473.1	313.4
+Stock Issuance	37.2	51.1	79.3	114.3	109.3
-Stock Purchases	123.1	113.9	257.0	250.7	864.0
-Dividends Paid	0.0	0.0	0.0	6.0	0.0
-Other Items ^a	1.2	-2.7	3.7	-248.7	13.5
Net Cash Flow	99.0	191.8	126.2	579.4	-454.9
Cash and Short-Term Investments	322.9	514.7	640.9	1,220.4	765.5
Common Shares Outstanding ^b (Split-Adj)	261.5	271.7	282.8	330.2	320.1
Share Price, Fiscal Year Close (Split-Adj)	\$15.84	\$33.25	\$67.94	\$44.21	\$38.33
MICROSOFT (June Fiscal Year)					
Sales	14,484.0	19,747.0	22,956.0	25,296.0	28,365.0
Income Before Taxes	7,117.0	11,891.0	14,275.0	11,525.0	11,513.0
-GAAP Income Taxes	2,627.0	4,106.0	4,854.0	3,804.0	3,684.0
Net Income	4,490.0	7,785.0	9,421.0	7,721.0	7,829.0
+Depreciation and Amortization	528.0	483.0	668.0	1,536.0	1,014.0
-Capital Expenditure	656.0	583.0	879.0	1,103.0	770.0
-Working Capital Investments	-335.0	1,470.0	182.0	-1,661.0	-3,882.0
+Tax Adjustment	1,527.0	3,232.0	4,054.0	2,504.0	1,784.0
Operating Cash Flow	6,224.0	9,447.0	13,082.0	12,319.0	13,739.0
+Stock Issuance	2,512.0	4,457.0	2,245.0	1,620.0	1,497.0
-Stock Purchases	2,468.0	2,950.0	4,896.0	6,074.0	6,069.0
-Dividends Paid	28.0	28.0	13.0	0.0	0.0
-Other Items ^a	1,279.0	7,617.0	3,856.0	63.0	2,115.0
Net Cash Flow	4,961.0	3,309.0	6,562.0	7,802.0	7,052.0
Cash and Short-Term Investments	13,927.0	17,236.0	23,798.0	31,600.0	38,652.0
Common Shares Outstanding (Split-Adj)	4,940.0	5,109.0	5,283.0	5,383.0	5,359.0
Share Price, Fiscal Year Close (Split-Adj)	\$54.19	\$90.19	\$80.00	\$73.00	\$54.70

Source: Adapted from Compustat.

^aOther Items includes long-term investments and acquisitions, extraordinary items, and other adjustments to net income.^bOn April 11, 2001, Maxim acquired Dallas Semiconductor Corporation with 41 million shares of stock.